

# Ask the Agent Nicely: Two Authorization Bypasses in n8n AI Agents

**Ask the Agent Nicely: Two Authorization Bypasses in n8n AI Agents** - Antonio De Turris, Antonio De Turris.

- Published: 2026-09-14
- Original: <<https://deturris.io/posts/n8n-ai-agents-authorization-bypasses/>>
- Preserved from: <https://deturris.io/posts/n8n-ai-agents-authorization-bypasses/> (live) on 2026-10-01
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## Content

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## TL;DR

As part of my vulnerability research into n8n’s AI Agents feature, I identified the following two vulnerabilities:

- **CVE-2026-65015**: a user with the read-only Project Viewer role can instruct an AI agent to execute arbitrary n8n nodes on their behalf, using the project’s credentials. This allows a supposedly read-only role to perform unauthorized actions, including exfiltrating project credentials and, under specific configurations, executing commands on the host running n8n.
- **CVE-2026-59207**: the “Allowed HTTP Request Domains” restriction configured on a credential is enforced by the application’s other functionalities that send HTTP requests,

but not by the agent's MCP client. A user authorized to use a credential but not to read it can point an MCP server at a host they control and exfiltrate the credential.

## Introduction

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n8n organizes work into [projects](#). A project groups workflows and credentials, where credentials hold the authentication material a workflow needs to reach an external service, for example when making an authenticated HTTP request.

Projects come in two kinds. Every user has a personal project, which is what most Community instances rely on: the resources belong to that user, and access is granted by sharing individual items with other users. [Team projects](#) instead have members, and each member is assigned a [project role](#) that governs what they can do with everything the project contains. Enterprise plans make more of these roles available, including the read-only [Project Viewer](#) role relevant to the two vulnerabilities discussed below.

Once a credential is created, its plaintext value can no longer be viewed through the n8n platform, including by the user who created it. Users with the appropriate permissions can modify a credential and, for HTTP credential types, restrict the domains to which the secret can be sent. This is intended to prevent unauthorized users from using a credential to send its secret to a server they control and recovering its value.

Over the past few months, n8n has added AI features that let users perform operations through the platform by chatting with an agent, instead of having to manually build a workflow.

Adding AI features to modern web applications can introduce new attack surface and, consequently, new vulnerabilities. The vulnerabilities discussed in this post are a concrete example of this.

## CVE-2026-65015: read-only Project Viewer can execute arbitrary nodes via `run_node_tool`

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**Affected versions:** all versions before 2.29.8, plus 2.30.0. Fixed in 2.29.8 and 2.30.1.

**Attacker prerequisites:** one authenticated user, member of a project, holding the Project Viewer role within the project.

Project Viewer is the read-only role within an n8n project. A Project Viewer can browse workflows, see which credentials are available to the project, and interact with the project's AI agents. They cannot execute workflows, use the nodes that make up those workflows, or access the plaintext value of project credentials.

The agent runtime introduces a different way to execute individual nodes through a built-in tool called `run_node_tool`. The tool runs n8n tool nodes server-side, with a large part of the node catalogue available to it, including the HTTP Request node. While a Viewer cannot execute a workflow, they can ask the agent to execute an individual node and provide arbitrary parameters for it.

The agent therefore opens a path around the privilege boundary. The attacker does not need permission to execute a workflow or read a credential. They only need to interact with the agent and have it invoke the desired node on their behalf.

In the following example, we simulate an attacker using an account that belongs to the TeamProj project and has the Project Viewer role. The user can see that a Header Auth credential named `PROD-SECRET` exists and obtain its ID (`VUQZnF9lyzE2rnPf`), but cannot edit its configuration or read its value. The user also cannot create workflows or agents, as shown by the disabled buttons in the screenshot below.

[\[image: Attacker holding Project Viewer role can see the list of credentials\]](#)

*Attacker holding Project Viewer role can see the list of credentials*

n8n allows you to configure an agent to use any model exposed through an OpenAI-compatible endpoint. For this proof of concept, the agent uses Qwen2.5-3B-Instruct, served locally through Ollama. Chatting with the agent is one of the few actions available to the Project Viewer, and it is all the attacker needs to trigger the vulnerability. As shown in the image below, the attacker sends a message instructing the agent to execute an HTTP Request node against an attacker-controlled Burp Collaborator URL, using the `PROD-SECRET` credential for authentication.

[\[image: The Viewer's chat message producing the tool call\]](#)

*The Viewer's chat message producing the tool call*

In response to the user's message, the agent invokes `run_node_tool` to execute the HTTP Request node. This allows the Project Viewer to make an HTTP request without having the permissions required to execute the node directly. Since the request is sent to an endpoint controlled by the attacker, the attacker can capture the request and recover the credential value by reading it from the `X-API-Key` header.

[\[image: The credential intercepted by Burp Collaborator\]](#)

*The credential intercepted by Burp Collaborator*

The root cause can be traced to two main sections of the 2.26.4 source code.

The first is the scope protecting the agent chat endpoint, in [agents.controller.ts#L560-L562](#):

```
// packages/cli/src/modules/agents/agents.controller.ts
@Post('/:agentId/chat', { useTemplates: true })
@ProjectScope('agent:execute')
async chat(...) { ... }
```

Any user with the `agent:execute` scope can chat with an agent and, through the agent, invoke `run_node_tool`. The project role definitions show that the read-only Project Viewer is granted this scope, while it does not have `workflow:execute`, as shown in [project-scopes.e.ts#L165-L185](#):

```
// packages/@n8n/permissions/src/roles/scopes/project-scopes.ee.ts
export const PROJECT_VIEWER_SCOPES: Scope[] = [
  'agent:read',
  'agent:list',
  'agent:execute', // <- chat with agents, and thus run_node_tool
  // ...
  'workflow:execute-chat',
  // ...
];
```

Another node that can be abused through this issue is [Execute Command](#), which runs shell commands directly on the server hosting n8n. The node is disabled by default from n8n 2.0 and is not available on n8n Cloud, but an administrator of a self-hosted instance can enable it. When enabled, the same attack chain allows a user holding the Project Viewer role to execute commands on the n8n host.

The video below demonstrates this by showing a user holding the Project Viewer role, in the same project, asking the agent to run the Execute Command node and obtaining a reverse shell on the Docker container running the n8n instance.

Your browser does not support the video tag. [Download the recording.](#)

*Project Viewer to reverse shell on the n8n host through the agent chat*

Official advisory: [GHSA-x5vx-c2c8-m3w9](#)

## CVE-2026-59207: the MCP connector ignores “Allowed HTTP Request Domains”

**Affected versions:** all versions before 2.27.4, plus 2.28.0. Fixed in 2.27.4 and 2.28.1.

**Attacker prerequisites:** one authenticated `global:member` user with use-only access to a shared credential.

n8n allows the owner of a credential to control which domains the credential can be sent to, through the [Allowed HTTP Request Domains](#) setting. The setting has three modes: `all`, `domains`, and `none`. With `none`, the credential cannot be sent in an HTTP request at all.

This restriction is what makes it possible to safely share a credential for use without exposing its value. A user can be allowed to use the credential, while being prevented from sending it to an arbitrary destination they control and recovering the secret from the request.

The HTTP Request node and n8n’s declarative routing engine both resolve the credential’s allowed domains and attach them to the outgoing request, and the shared HTTP request layer rejects any URL that falls outside them, redirects included. The screenshot below shows an attempt to use the `PROD-SECRET` credential with an HTTP Request node being blocked by n8n. The credential is configured with `Allowed HTTP Request Domains = None`.

[image: The HTTP Request node refusing to send the restricted credential]

*The HTTP Request node refusing to send the restricted credential*

The issue can be exploited entirely through the MCP configuration. A user with use-only access to the shared `PROD-SECRET` credential can configure an MCP server on a host they control and select the credential for authentication. They do not need to know the credential value or send a specially crafted message to the agent.

In the following example, we simulate an attacker using an account the credential has been shared with. From their own session, the user can see that `PROD-SECRET` exists and obtain its ID, but the credential is marked read only and belongs to another user, so its value is never displayed to them.

[image: The shared PROD-SECRET credential, read only, in the attacker's session]

*The shared PROD-SECRET credential, read only, in the attacker's session*

The attacker then creates an agent and adds an MCP server to its configuration. The server URL points to a Burp Collaborator endpoint they control, the authentication method is set to header authentication, and `PROD-SECRET` is selected as the credential to use.

[image: The agent configuration pointing an MCP server at an attacker-controlled URL]

*The agent configuration pointing an MCP server at an attacker-controlled URL*

Sending any message that causes the agent to reach the MCP server is enough to trigger the connection. The agent reports that the connection failed, because the attacker's endpoint does not implement the MCP protocol.

[image: The agent chat reporting a failed MCP connection]

*The agent chat reporting a failed MCP connection*

However, the failure is only visible on the n8n side. The request has already been sent, and it carries the plaintext credential in the `X-Api-Key` header, where the attacker can read it.

[image: The credential intercepted by Burp Collaborator in the X-Api-Key header]

*The credential intercepted by Burp Collaborator in the X-Api-Key header*

The root cause can be found in the MCP client implementation of the 2.26.0 source code. When building the client, n8n resolves the selected credential and converts it into HTTP headers, in `mcp-client-factory.ts#L43-L69`:

```
// packages/cli/src/modules/agents/json-config/mcp-client-factory.ts
headerAuth -> { [name]: value }
bearerAuth -> { Authorization: `Bearer ${token}` }
```

Unlike the other HTTP request paths, the MCP client does not check the credential's allowed domains before sending these headers. The agents module does not call `getCredentialAllowedDomains`, `isDomainAllowed`, or any equivalent check before connecting to the configured MCP server.

The result is a complete bypass of the intended credential restrictions: the user can provide an arbitrary destination, have the agent attach the shared credential to the request, and recover the secret from a server they control.

Official advisory: [GHSA-h44j-f5r5-ph73](#)

## Closing

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In this post, we looked at two authorization bypasses in n8n's AI features. The first allows a low-privileged user to execute arbitrary nodes, which is enough to use and exfiltrate project credentials they are not allowed to read and, when a node such as Execute Command is enabled, to run commands on the n8n host. The second allows a user to bypass the domain restriction configured on a credential and exfiltrate its plaintext value to a server they control.

Both issues came from the same underlying problem: security checks enforced by the application's existing functionality were missing from the new AI-driven paths.

Adding AI to an application also adds new ways for users to reach existing functionality. Those paths need to enforce the same authorization boundaries as the rest of the application, regardless of whether an action is triggered directly by a user or indirectly through an agent.

## Disclosure Timeline

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- 2026-06-15: CVE-2026-59207 reported to n8n
- 2026-06-16: CVE-2026-65015 reported to n8n
- 2026-06-24: n8n confirmed CVE-2026-59207, released the fix in 2.27.4 and 2.28.1, and published [GHSA-h44j-f5r5-ph73](#)
- 2026-07-08: n8n confirmed CVE-2026-65015, released the fix in 2.29.8 and 2.30.1, and published [GHSA-x5vx-c2c8-m3w9](#)

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